Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Please use A4 paper and write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules. Prepare a single solution file.
- Start working right away. There is no late submission form. If you miss the deadline, you need to use the make-up assignment to cover up the marks.

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1. In the classes, we discussed three forms of floating number representations as shown below,

$$\text{Standard Form} \quad : \quad F = \pm(0.d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \;, \quad (d_1 \neq 0) \tag{1}$$
$$\text{IEEE Normalized Form} \quad : \quad F = \pm(0.1 d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \;, \tag{2}$$
$$\text{IEEE Denormalized Form} \quad : \quad F = \pm(1.d_1 d_2 d_3 \cdots d_m)_\beta \, \beta^e \;, \tag{3}$$

   where $d_i, \beta, e \in \mathbb{Z}$, $0 \leq d_i \leq \beta - 1$ and $e_{\min} \leq e \leq e_{\max}$. Now, let's take, $\beta = 2$, $m = 5$ and $-2 \leq e \leq 5$. Based on these, answer the following:

   (a) (6 marks) What are the maximum numbers that can be stored in the system by these three forms defined above (express your answer in decimal values)?

   (b) (6 marks) What are the non-negative minimum numbers that can be stored in the system by the three forms defined above (express your answer in decimal values)?

   (c) (6 marks) Including negative numbers, what range of the floating numbers in these three representations are considered as ZERO and $\pm\infty$ because of the underflow and overflow respectively.

2. Consider the quadratic equation, $2x^2 - 70x + 3$. Keep up to 6 significant figures in all calculations below:

   (a) (3 marks) Evaluate the roots using the standard formula to obtain a solution of the quadratic equation.

   (b) (3 marks) Write down the fundamental properties the solutions of the given quadratic equation must satisfy. Does your solutions satisfy these properties? If not, compute the correct solutions.

3. Consider the function, $f(x) = \cos(x) + \sin(x)$. Up to 5 significant figures, $f(0.15) = 1.1382$. Now answer the following:

   (a) (3 marks) Express $f(x)$ as a Taylor series, and identify the Taylor coefficients a_0, a_1, a_2, a_3 a_4 and a_5.

   (b) (3 marks) What would be the degree of the interpolation polynomial if we would like to find $f(0.15)$ by Taylor series expansion.

   (c) (3 marks) Compute the upperbound of the interpolation error at $x = 0.15$ and for $n = 3$.

4. Consider the function $f(x) = \sin(x) + \cos(x)$ at the nodes $\{-\pi, \, -\pi/2, \, 0\}$. Answer the following:

   (a) (3 marks) Write down the Vandermonde matrix V, and find the interpolation polynomial using V.

   (b) (4 marks) Evaluate the Lagrange bases for the given nodes, and find the Lagrange interpolation polynomial.

Motto: Mathematics is NOT difficult, but what is difficult is to believe that mathematics is NOT difficult.